

Compound Helicopter - Assignment 1

NAMES: Kshitij Kumar Pradhan, Rajdeep Paul , Anupchand Yadav, Binay Kumar Shaw

ROLL NUMBERS: 22b4215,23b0035, 22b0023,22b0020

Team Member Contribution

Sr.No	Roll Number	Name	Contribution Level (0 - 5)	Specifics of contribution
1	22b4215	Kshitij Kumar Pradhan	5	Completed all the parts of the assignment
2	23b0035	Rajdeep Paul	0	Not responsive
3	22b0023	Anupchand Yadav	0	Not responsive
4	22b0020	Binay Kumar Shaw	0	Not responsive

Section 1:

Assumptions & Data

1.1 Physics Assumptions/Data

- The simulation is based on Blade Element Momentum Theory (BEMT).
- Forward flight inflow is approximated using Glauert's momentum theory.
- Blade flapping dynamics are simplified to the first harmonic (coning and cyclic flapping).

1.2 Environmental Assumptions/Data

- All calculations use the International Standard Atmosphere (ISA) to model air density, temperature, and pressure.
- Wind is assumed to be steady, with its effects included by adjusting the helicopter's airspeed.

1.3 Vehicle Assumptions/Data

- Rotor blades are modeled as rigid structures with a constant chord and linear twist.
- Fuselage drag is simplified to an equivalent flat plate area.
- All specific design parameters (weight, dimensions, etc.) are loaded from external JSON configuration files.

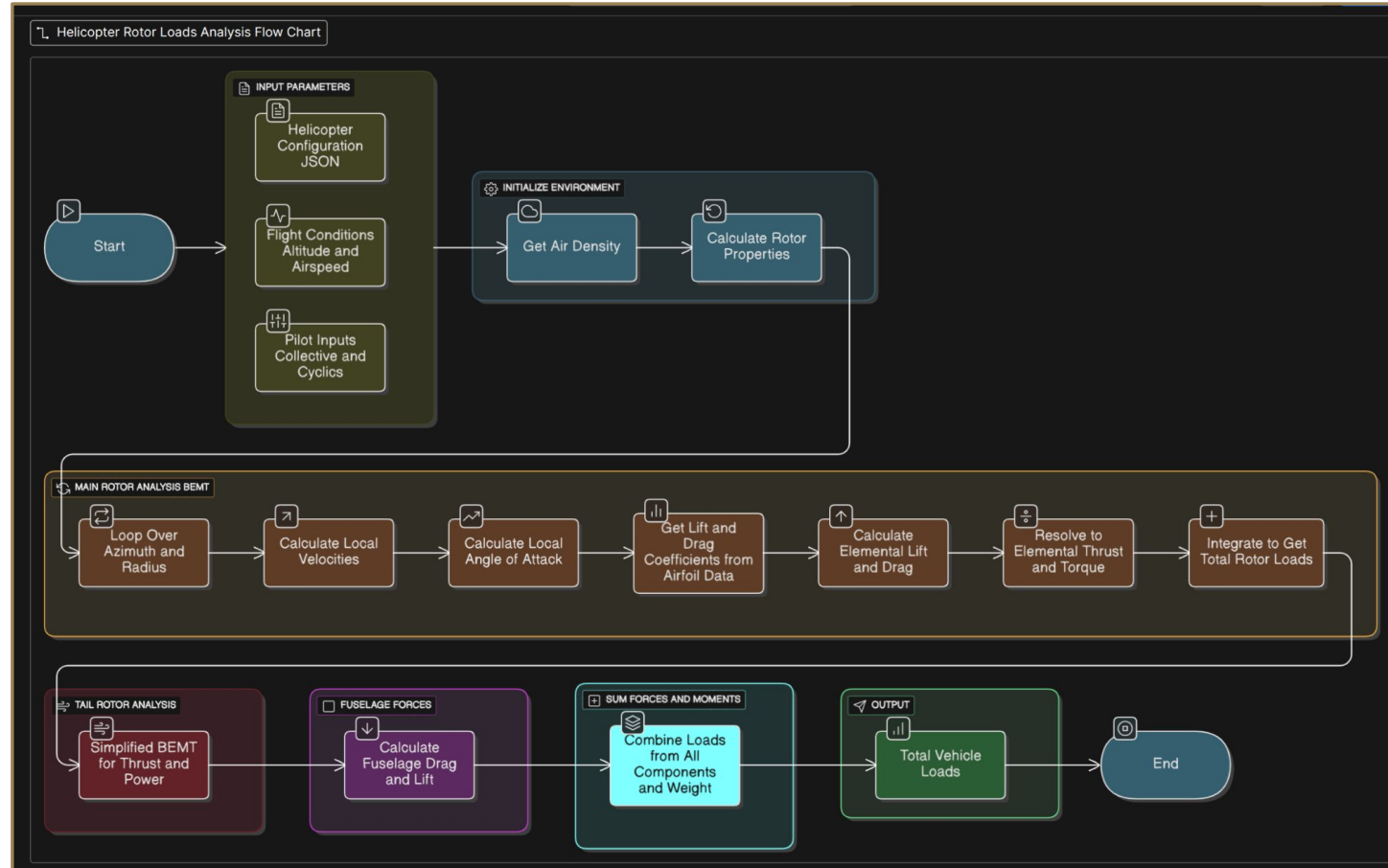
1.4 Flight Condition Assumptions/Data

- The primary analysis focuses on steady, level, and unaccelerated flight.
- The trim and performance analyses are conducted at a constant altitude of 2000 meters AMSL.

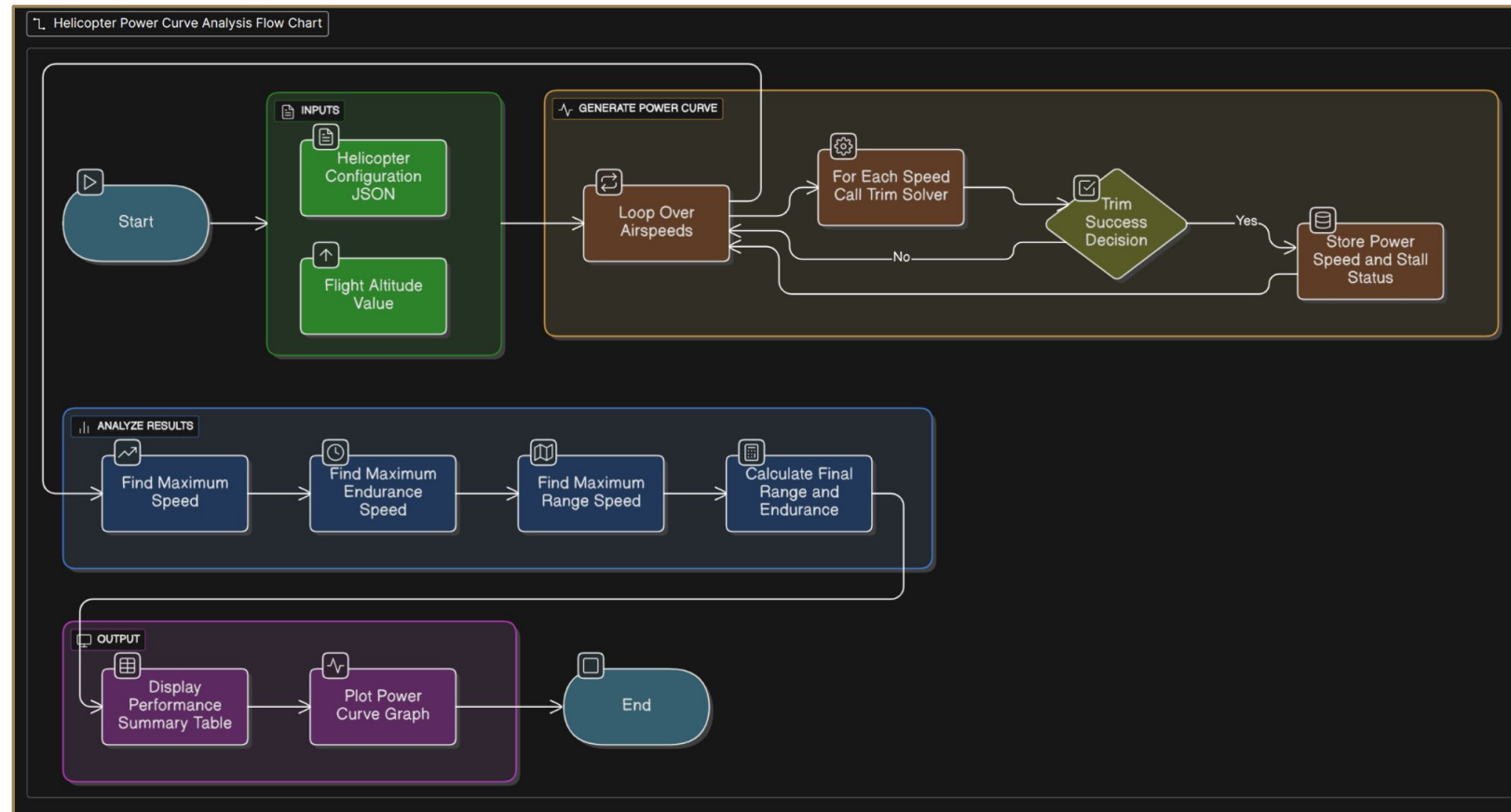
Section 2:

Algorithm/Logic Flow Diagrams

2.1: Working/Algorithm/Logic Flow Diagram of the Performance Estimator Tool



2.2: Working/Algorithm/Logic Flow Diagram of the Mission Planner

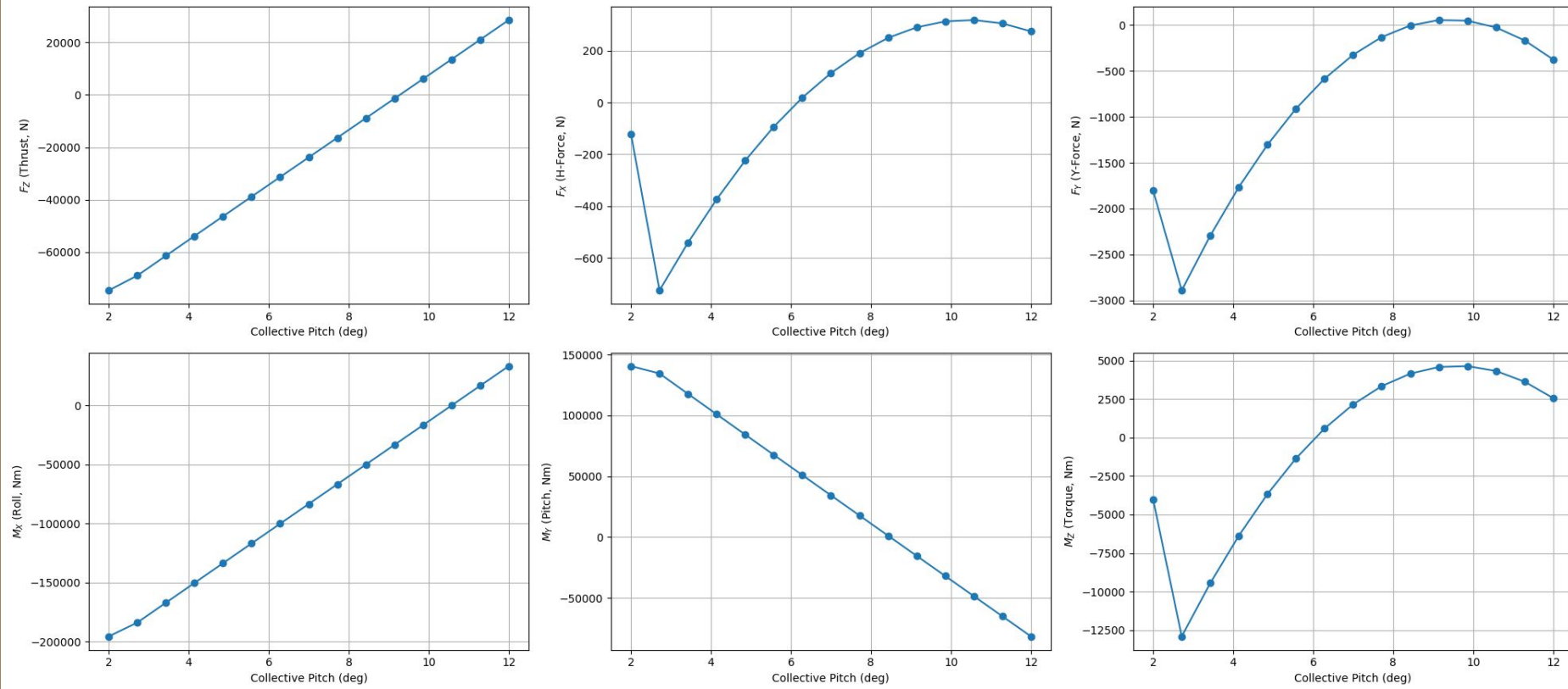


Section 3:

Pilot Input Tests and Observations

3.1: Plot of F_x , F_y , F_z , M_x , M_y , M_z with changes in pilot inputs

Forces & Moments vs. Main Rotor Collective (200 km/h @ 2000m)



3.2: Observation and reasoning for behaviour of F_x , F_y , F_z , M_x , M_y , M_z .

As collective pitch increases, the angle of attack on all rotor blades rises, leading to a predictable and near-linear increase in the magnitude of Thrust (F_z) and the required drive Torque (M_z). The Pitching (M_y) and Rolling (M_x) moments also change linearly, as the higher overall lift amplifies the effects of the fixed cyclic inputs required for 200 km/h flight. The in-plane H-Force (F_x) and Side Force (F_y) show non-linear trends due to the complex redistribution of blade profile and induced drag across the advancing and retreating sides of the rotor disk.

Section 4:

Trim Settings

4.1: Trim settings and resultant forces and moments

--- Trim Results for 200 km/h at 2000m ---

Input		Resultant	
theta_0	= 17.74 deg	Fx (Vehicle)	= -927.37 N
theta_1s	= -12.48 deg	Fy (Vehicle)	= -3132.50 N
theta_1c	= 7.63 deg	Fz (Vehicle)	= 51.32 N
theta_0_t	= 228.29 deg	Mx (Vehicle)	= -105674.29 Nm
alpha_tpp	= 3.40 deg	My (Vehicle)	= -38.50 Nm
beta_0	= 24.89 deg	Mz (Vehicle)	= 225.23 Nm

Section 5:

Mission Planner Test

Findings for compound helicopter in trim conditions (at 2000m AMSL)

```
--- Mission Performance Findings at 2000 m AMSL ---  
+-----+  
| Metric | Value |  
+-----+  
| 4.1 Max Speed (Blade Stall) | 61.38 km/h |  
| 4.2 Max Speed (Power Requirement) | 20.00 km/h |  
| 4.3 Maximum Range | 79.19 km |  
| 4.4 Maximum Endurance | 1.29 hours |  
+-----+
```

--- Mission Performance Findings at 2000 m AMSL ---	
Metric	Value
4.1 Max Speed (Blade Stall)	61.38 km/h
4.2 Max Speed (Power Requirement)	20.00 km/h
4.3 Maximum Range	79.19 km
4.4 Maximum Endurance	1.29 hours

Acknowledgement

I would like to extend my sincere gratitude to our instructor Prof. Dhwanil Shukla for their comprehensive lectures and for providing the course materials that formed the theoretical basis of this project along with the constant support of the teaching assistants .

The development of the performance estimation tool and mission planner would not have been possible without the foundational knowledge imparted through the course. The implementation of the computational models relied heavily on the Python programming language and its scientific computing libraries, to solve the aerodynamic and trim equations.

Finally, I wish to acknowledge the authors of the key reference texts, whose work provided invaluable guidance and deeper insight into the principles of rotary-wing aerodynamics.

References

1. Leishman, J. Gordon. Principles of Helicopter Aerodynamics, 2nd Edition, 2006.
2. Prouty, Ray W. Helicopter Performance, Stability, and Control, 2nd Edition, 2002.
3. Johnson, Wayne. Helicopter Theory, 1994.
4. Glauert, H. "A General Theory of the Autogyro," British A.R.C. R. & M. No. 1111, 1926.
5. Knight, M., & Hefner, R. A. "Static Thrust Analysis of the Lifting Airscrew," NACA TN 626, 1937.
6. UIUC Airfoil Data Site: <https://m-selig.ae.illinois.edu/ads.html>
7. FAA Helicopter Flying Handbook:
https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/helicopter_flying_handbook